Predicting Demand & Detecting Anomalies in Urban Bike-Share Systems

A Data Mining Project Proposal

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AGENDA

Outline

- Problem & Motivation
- Related Work
- Proposed Approach: Data, Tools & Techniques
- Evaluation Plan
- Timeline
- Questions & Discussion

PROBLEM & MOTIVATION

Bike-share systems struggle with demand imbalance

- Stations empty near business districts during AM commute; overflow near residential areas in PM
- Manual truck-based "rebalancing" is costly and reactive
- Anomalous trips (theft, GPS errors, malfunction) go undetected without systematic monitoring

> **Research question:** Can historical trip, weather, and calendar data predict near-term station demand well enough to guide rebalancing — and automatically flag anomalous trips worth investigating?

RELATED WORK

Prior work in bike-share analytics

- **Froehlich et al. (2009)** — station usage shows strong, predictable spatio-temporal regularities
- **O'Mahony & Shmoys (2015)** — Citi Bike rebalancing as a data-driven optimization problem
- **Li et al. (2015)** — clustering stations before demand prediction improves accuracy
- **Chen et al. (2016)** — combines clustering + classification for over-demand prediction

> This project extends these approaches by adding an explicit outlier-detection and frequent-pattern-mining layer.

PROPOSED APPROACH

Data & Tools

Data

- Citi Bike NYC trip histories (public CSV)
- NOAA weather data (temp, precipitation)
- Calendar features (day-of-week, holidays)

Tools

- Python: pandas, scikit-learn
- mlxtend (FP-Growth / association rules)
- PyOD (outlier detection)
- SQLite / Parquet data warehouse

PROPOSED APPROACH

Techniques Applied Across the Pipeline

Stage	Technique	Purpose
Preprocessing / Warehousing	Cleaning + star schema	Reliable, query-ready trip data
Clustering	k-means / DBSCAN	Station usage-profile groups
Classification	Random Forest / GBM	Predict high/low demand station-hours
Frequent patterns	FP-Growth	Common origin-destination sequences
Outlier detection	Isolation Forest	Flag anomalous trips

EVALUATION PLAN

How we'll measure success

- **Classification:** F1-score vs. historical-average baseline (target: +10%)
- **Clustering:** Silhouette score + interpretability of station groups
- **Frequent patterns:** Support / confidence / lift, manually reviewed for plausibility
- **Outlier detection:** Manual review precision on flagged trip sample

TIMELINE

Project Phases (8 weeks)

Phase	Weeks	Deliverable
Data acquisition & preprocessing	1–2	Clean dataset + warehouse
Exploratory analysis & clustering	3	Station clusters
Classification & pattern mining	4–5	Demand classifier + rules
Outlier detection & evaluation	6	Anomaly results + metrics
Final report & presentation	7–8	Report, slides, code

Questions & Discussion

Thank you!